

TECHBIG SOLUTIONS

SAP BASIS – S/4HANA & AWS Cloud

Complete Professional Training Syllabus

Ambattur, Chennai

Course Objective

A comprehensive technical administration syllabus combining SAP Basis for SAP S/4HANA with SAP HANA administration and SAP workloads on Amazon Web Services (AWS). The program covers installation, configuration, users and authorizations, transports, jobs, monitoring, performance, backup/recovery, HA/DR, upgrades, security, Fiori administration, HANA operations, AWS infrastructure, SAP on AWS architecture, migration and real-time production support.

The AWS portion follows AWS's SAP guidance covering EC2, EBS, EFS, S3, VPC, networking, certified instance types, SAP HANA on AWS, high availability/disaster recovery, monitoring and migration planning.

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1. SAP BASIS Fundamentals

- SAP Basis role and responsibilities
- SAP S/4HANA technical architecture
- Application server, database and presentation layers
- SAP system landscape: DEV, QAS, PRD
- SID, client, instance and system number
- SAP profiles and parameter concepts
- Work processes and dispatcher
- Message server and enqueue server
- SAP services and start/stop architecture

2. SAP S/4HANA Installation & System Setup

- Installation planning and prerequisites
- Software Provisioning Manager overview
- SAP S/4HANA installation architecture
- SAP application server installation
- SAP HANA database connectivity
- Post-installation configuration
- License and system checks
- System copy/refresh overview
- Kernel and SAP Host Agent
- SAP Notes and support packages

3. SAP HANA Administration for Basis

- SAP HANA architecture
- Tenant and system database concepts
- HANA services and processes
- HANA cockpit overview
- HANA Studio/administration tooling overview
- HANA users and roles
- HANA memory and CPU concepts
- HANA configuration parameters
- HANA backup and recovery
- HANA system replication
- HANA performance monitoring

4. SAP GUI, Fiori & User Administration

- SAP GUI configuration
- SAP Logon and connections
- Client concepts

- User creation and maintenance
- User types
- Roles and profiles
- Authorization concepts
- Fiori Launchpad architecture
- Fiori catalogs, spaces and pages overview
- Fiori services and troubleshooting

5. SAP Security & Authorizations

- Authorization objects
- PFCG roles
- Single and composite roles
- Role menus
- User-role assignment
- SU01/SU10 administration
- SU24 concepts
- Trace and authorization analysis
- S_TCODE and critical authorizations
- Background and technical users
- Security hardening and audit basics

6. Transport Management System (TMS)

- Transport landscape design
- Domain controller
- Transport routes
- STMS configuration
- Workbench and Customizing requests
- Import queues
- Import history and logs
- Transport errors
- Transport of copies
- ChaRM/SolMan overview
- Emergency transport procedures

7. Background Jobs & Batch Administration

- SM36/SM37
- Job scheduling
- Periodic jobs
- Job variants
- Job classes and priorities
- Job logs and spool

- Cancelled job analysis
- External commands
- Event-based scheduling
- Housekeeping and job monitoring

8. Spool, Printing & Output Administration

- SPAD concepts
- Output devices
- Device types
- Spool requests
- Output requests
- Printer configuration
- Spool housekeeping
- Email output overview
- Troubleshooting printing issues

9. System Monitoring & Troubleshooting

- SM50/SM66 work process monitoring
- SM51 application server overview
- SM21 system log
- ST22 ABAP dumps
- SM13 update errors
- SM12 lock entries
- SM04/AL08 user monitoring
- ST02 buffers
- ST03N workload analysis
- ST06 OS monitoring
- Root-cause analysis methodology

10. SAP Performance Tuning

- Work process bottlenecks
- CPU and memory analysis
- Database performance
- HANA performance overview
- SQL analysis
- ST05 SQL trace
- ST12 combined trace
- SAT overview
- Buffer and cache concepts
- Expensive statements
- Application-server sizing concepts

11. SAP Kernel & Software Maintenance

- SAP kernel architecture
- Kernel patching
- SAPEXE/SAPEXEDB concepts
- SAPCAR
- SAP Host Agent
- Support Packages
- Support Package Stack
- SAP Notes implementation overview
- SUM overview
- Maintenance Planner concept

12. SAP System Copy, Refresh & Client Administration

- Client concepts
- Local and remote client copy
- Client export/import
- System copy overview
- Homogeneous/heterogeneous copy concepts
- Post-copy automation overview
- BDLS concept
- Logical system configuration
- Client refresh planning
- Validation after refresh

13. Database Backup & Recovery

- HANA backup types
- Data backup
- Log backup
- Backup destinations
- Backup validation
- Recovery scenarios
- Point-in-time recovery
- Backup retention
- Recovery testing
- AWS storage considerations

14. High Availability & Disaster Recovery

- HA concepts and objectives
- RTO and RPO
- ASCS/ERS architecture

- Enqueue Replication Server
- Cluster concepts
- HANA System Replication
- Primary/secondary HANA
- Synchronous/asynchronous replication
- Multi-AZ architecture
- DR runbooks and testing

15. SAP Fiori Administration

- Fiori architecture
- Frontend/backend concepts
- Embedded deployment overview
- Fiori Launchpad configuration
- ICF services
- OData services
- Catalogs, spaces and pages
- System aliases
- Gateway troubleshooting
- Cache cleanup and invalidation
- Fiori performance and security

16. SAP Gateway & OData Administration

- SAP Gateway architecture
- OData service activation
- SICF concepts
- /IWFND and /IWBEP monitoring
- Service registration
- Metadata and cache troubleshooting
- Authorization issues
- Error logs
- Backend connectivity

17. SAP HANA Security & Operations

- HANA users and authentication
- HANA roles
- Privilege types
- Auditing
- Encryption overview
- HANA cockpit monitoring
- Memory management
- Savepoints and log segments

- HANA alerts
- Tenant management

18. SAP Solution Manager / ALM Overview

- Solution Manager architecture
- System monitoring
- Technical monitoring
- EarlyWatch concepts
- ITSM overview
- ChaRM overview
- SAP Cloud ALM overview
- Operations and incident management

19. SAP S/4HANA Upgrade & Conversion

- S/4HANA upgrade concepts
- SUM with Database Migration Option overview
- Software Update Manager
- Maintenance Planner
- Simplification Item concept
- Readiness checks
- Custom code considerations
- Downtime optimization
- Post-upgrade validation
- Fallback and contingency planning

20. AWS Cloud Fundamentals for SAP Basis

- AWS global infrastructure
- Regions and Availability Zones
- AWS accounts and IAM
- VPC and subnets
- Route tables
- Internet/NAT gateways
- Security Groups and NACLs
- EC2 fundamentals
- EBS volumes
- EFS and S3
- CloudWatch overview

21. SAP on AWS Architecture

- SAP All-on-AWS architecture
- Hybrid SAP architecture

- Network connectivity
- VPN and AWS Direct Connect
- SAProuter
- Private/public subnets
- Multi-AZ design
- SAP-certified EC2 instance types
- Storage architecture
- SAP HANA on AWS
- AWS shared responsibility model

22. EC2 for SAP S/4HANA

- EC2 instance selection
- SAP-certified instances
- AMI concepts
- EBS volume types
- EBS performance and sizing
- Instance lifecycle
- Elastic IP concepts
- Security groups
- OS administration
- SAP application server on EC2
- HANA database on EC2

23. AWS Networking for SAP

- VPC architecture
- CIDR and subnet design
- Route tables
- Internet Gateway
- NAT Gateway
- Transit Gateway overview
- VPN
- Direct Connect
- DNS and Route 53 overview
- Network security
- Connectivity troubleshooting

24. AWS Storage for SAP

- EBS architecture
- Provisioned IOPS considerations
- EFS use cases
- S3 storage

- Backup storage
- Snapshot concepts
- S3 lifecycle policies
- Storage performance
- SAP HANA persistence design

25. AWS IAM & Security for SAP

- IAM users, groups and roles
- Policies
- Least privilege
- IAM roles for EC2
- KMS and encryption
- Secrets management overview
- Security Groups
- CloudTrail
- AWS Config overview
- SAP application security on AWS

26. SAP HANA on AWS Operations

- HANA deployment planning
- Certified EC2 instances
- Operating systems for HANA
- HANA storage layout
- HANA installation overview
- HANA backup to AWS storage
- HANA System Replication
- Monitoring
- AWS Data Provider for SAP
- HANA troubleshooting

27. AWS High Availability & DR for SAP

- Multi-AZ design
- HANA System Replication
- ASCS/ERS HA
- Cluster solutions
- Failover planning
- Backup-based DR
- Cross-Region DR
- Amazon S3 replication
- RTO/RPO design
- DR testing and runbooks

28. SAP Backup, Monitoring & Operations on AWS

- AWS Backup concepts
- EBS snapshots
- S3 backup repositories
- CloudWatch metrics
- CloudWatch alarms
- AWS Data Provider for SAP
- SAP monitoring integration
- Log management
- Operational dashboards
- Alerting and incident response

29. SAP Migration to AWS

- Migration assessment
- SAP workload discovery
- AWS sizing
- Network planning
- Homogeneous migration overview
- Database migration options
- Backup/restore migration
- Downtime planning
- Cutover strategy
- Validation and rollback
- Post-migration optimization

30. AWS Automation for SAP

- AWS Systems Manager overview
- SSM automation
- Patch management
- EC2 automation
- CloudFormation overview
- Infrastructure as Code concepts
- AWS Launch Wizard for SAP overview
- Auto-recovery concepts
- Operational automation
- Cost optimization

31. SAP on AWS Cost & Capacity Management

- EC2 pricing models
- On-Demand and Reserved capacity concepts

- Savings Plans overview
- Storage cost optimization
- Right-sizing
- Non-production scheduling
- Tagging strategy
- Cost allocation
- Capacity planning

32. Real-Time End-to-End Projects

- Project 1: Install and configure S/4HANA landscape
- Project 2: Configure TMS DEV-QAS-PRD
- Project 3: User/role/Fiori administration
- Project 4: HANA backup and point-in-time recovery
- Project 5: HANA System Replication HA
- Project 6: Build SAP S/4HANA on AWS architecture
- Project 7: Deploy SAP application and HANA on AWS EC2
- Project 8: Configure VPC, security and connectivity
- Project 9: Migrate a non-production SAP system to AWS
- Project 10: Production incident, performance analysis and DR drill

33. Production Support & Interview Preparation

- Daily Basis checklist
- Incident and problem management
- SAP monitoring checklist
- HANA monitoring checklist
- AWS monitoring checklist
- Performance troubleshooting scenarios
- Transport and job failures
- User/Fiori issues
- Backup/restore incidents
- HA/DR incidents
- SAP on AWS architecture interview questions
- Real-time project discussion and mock interviews

Recommended Training Roadmap

Phase	Coverage	Outcome
1	Basis Foundation	Understand S/4HANA architecture and daily administration
2	HANA Administration	Operate, monitor, secure and recover HANA
3	S/4HANA Operations	TMS, jobs, security, Fiori, Gateway and performance
4	HA/DR	Design and operate resilient SAP landscapes
5	AWS Foundation	EC2, VPC, storage, IAM, monitoring and security
6	SAP on AWS	Deploy, operate, backup, migrate and optimize SAP
7	Projects	Implement complete S/4HANA on AWS scenarios
8	Support & Interview	Production troubleshooting and real-time interview preparation

Current Reference Alignment

AWS's SAP documentation specifically covers SAP on AWS planning, EC2 instance selection, networking, storage, operating systems, HANA deployment, monitoring, backup/recovery, high availability and disaster recovery. AWS also documents both all-on-AWS and hybrid SAP architectures. ■cite■turn0search0■turn0search2■turn0search7■

Note: AWS's former SAP on AWS Specialty certification was retired in 2024; this syllabus therefore focuses on practical SAP Basis/S/4HANA and SAP-on-AWS skills rather than treating that retired certification as a current target.

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